

Placement Operations Copilot

Decision-Support System for Placement Managers

Imarticus Data Science Internship – Cohort 6

1. Introduction

Placement teams often face challenges when deciding whether a candidate is ready for interviews, which roles best suit them, and what specific improvements are required before scheduling interviews. These decisions are usually based on fragmented data such as skill scores, interview feedback, and subjective judgment.

The **Placement Operations Copilot** is designed to address this problem by providing a **structured, explainable, and data-driven decision-support system**. Instead of automating decisions, the system assists placement managers by analyzing candidate data and presenting clear insights that help them make informed placement decisions.

The solution is built as a **local and offline system**, ensuring simplicity, transparency, and ease of evaluation without reliance on cloud services or external APIs.

2. Objective of the Project

The primary objective of this project is to design and implement an **agentic decision-support workflow** that helps placement managers:

- Evaluate candidate readiness for interviews
- Recommend suitable job roles based on skill matching
- Identify strengths and gaps from interview feedback
- Suggest actionable next steps and preparation timelines

The system focuses on **reasoning and explainability**, ensuring that every recommendation is backed by clear logic rather than black-box predictions.

3. System Overview

The Placement Operations Copilot works by combining **rule-based logic** with structured data analysis. Candidate skill scores (such as Python, SQL, Excel, Statistics, and Communication) are processed along with role requirements to determine how well a candidate fits specific job roles.

The system is composed of modular components, each responsible for a specific task:

- Readiness evaluation
- Role suitability analysis
- Feedback analysis
- Action planning

All insights are displayed using an interactive **Streamlit dashboard**, allowing placement managers to easily explore candidate profiles, compare roles, and review recommended actions.

4. Core Functional Workflows

4.1 Candidate Readiness Evaluation

The system evaluates each candidate's overall readiness by analyzing their average skill scores and key competencies. Based on predefined thresholds, candidates are classified into one of the following categories:

- **Ready**
- **Almost Ready**
- **Not Ready**

Along with the readiness status, the system provides clear reasoning explaining why a candidate falls into a particular category and highlights areas that require improvement.

4.2 Role Suitability Recommendation

For each candidate, the system evaluates suitability across multiple roles, including:

- Junior Data Analyst
- Data Analyst
- Business Analyst
- Data Engineer
- Data Scientist
- Machine Learning Intern

A **match percentage** is calculated for every role by comparing candidate skills with role-specific requirements. The system then:

- Ranks roles based on suitability
- Recommends the top 2–3 best-fit roles
- Highlights skill gaps preventing higher role alignment

This helps placement managers quickly identify where a candidate is most likely to succeed.

4.3 Interview Feedback Analysis

The feedback analysis module examines interview feedback and categorizes insights into:

- **Strengths** (areas where the candidate performs well)
- **Gaps** (areas requiring improvement)

By separating strengths and gaps, the system ensures feedback is constructive and actionable rather than generic.

4.4 Next Actions Summary

The final workflow combines all insights into a concise **Next Actions Summary**, which includes:

- Candidate readiness status
- Recommended roles
- Key improvement areas

- A suggested **7-day preparation timeline**

This summary is designed to support quick and confident decision-making by placement teams.

5. Multi-Candidate Support

The system supports **multiple candidates** using a single dataset. Candidates are loaded from a CSV file, and placement managers can switch between them using a dropdown in the dashboard.

Each candidate receives:

- A unique readiness status
- Personalized role recommendations
- Individual strengths, gaps, and action plans

This makes the solution scalable and closer to real-world placement scenarios.

6. Dashboard and User Interface

The Streamlit dashboard presents insights in a clean and intuitive layout, featuring:

- Candidate profile and skill-wise score table
- Readiness status with explanations
- Role recommendations with match percentages and progress bars
- Strengths and gaps comparison
- Clear action-oriented recommendations

The interface prioritizes clarity and usability, ensuring placement managers can quickly understand and act on the insights provided.

7. Technology Stack

- **Programming Language:** Python
- **Libraries:** Pandas, Streamlit
- **Data Formats:** CSV, JSON, TXT
- **Execution Mode:** Fully local and offline

No external APIs or cloud services are used, keeping the system simple and transparent.

8. Assessment Alignment

This project fully satisfies the assessment requirements by:

- Implementing an agentic, rule-based decision-support workflow
- Covering all mandatory workflows specified in the assessment
- Providing explainable logic for all recommendations
- Including code, dashboard, report, outputs, and documentation
- Running completely offline without external dependencies

9. Conclusion

The Placement Operations Copilot demonstrates how structured reasoning and data analysis can significantly improve placement decision-making. By focusing on explainability, modular design, and real-world applicability, the system provides placement managers with a reliable tool to assess candidates, recommend suitable roles, and plan interview readiness effectively.

This project reflects a practical understanding of data-driven decision support and aligns closely with real-world placement operations.